

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 37

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • The time taken for half of the atoms of a radioactive isotope to decay (activity halves) (1)	1
01.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $32 / 8 = 4$ half-lives (1) • $6400 / 2^4 = 400$ Bq (1)	2
01.3	<i>1 mark for each point.</i> • Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient (1) • Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low (1) • Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) (1)	3
01.4	<i>1 mark for each point.</i> • Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure (1) • Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) (1) • Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) (1)	3
01.5	<i>1 mark.</i> • A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
02.1	<i>1 mark for each point.</i> - The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence (1) - An individual event, or even several, does not show a trend; the change may be in the frequency of events which can only be assessed years later (1) - Changes occur at different time and spatial scales (local storm vs global trend); interconnected systems make attribution complex (1)	3
02.2	<i>1 mark for each point.</i> - Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) (1) - As the town grows around the weather station the warming recorded is partly local, not global; older equipment / sparse global coverage also reduce reliability (1)	2
02.3	<i>Up to 5 marks in total; award marks in proportion to the completeness of each point.</i> - Tree rings: width of each annual ring shows the growth rate, indicating temperature / rainfall in that year; rings easily dated by counting; limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions (up to 2) - Ice cores: annual snow layers compact into ice; air bubbles trapped in the ice give the CO₂ (and methane) concentration when trapped; oxygen isotope ratios indicate temperature; up to 800 000 years (Dome C); limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers; only polar regions (up to 2) - Others accepted: coral growth rings; pollen in lake sediments dated by radiocarbon (up to 2)	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises (2) - Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) (2) - Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises (2)	6
03.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Rice paddy fields (1) - Landfill sites (1) - Anaerobic sediments in reservoirs (1) - Livestock intestines (cattle) (1) - Manure / slurry stores (1)	3

Question	Marking guidance	Marks
03.3	1 mark. • More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
04.1	1 mark for each point. • Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) (1) • Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) (1)	2
04.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. • $(200 - 25) / 200 \times 100$ (1) • = 87.5% (1)	2
04.3	Any 4 from the points below - 1 mark each, maximum 4. • DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) (1) • Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths (1) • Higher doses caused sterility / death (1) • After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) (1) • Breeding success recovered so the population rose (1)	4
04.4	Any 2 from the points below - 1 mark each, maximum 2. • DDT persists in soil / sediments / food chains for many years so exposure continued (1) • Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly (1) • Other pressures (habitat loss, persecution) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
05.1	Correct answer only. • B - rivers are accessible over a large area with few natural contaminants (1)	1
05.2	1 mark for each point. • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
05.3	1 mark for each point. • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable (1) • Wild species have different ranges of tolerance so cress results cannot be applied directly (1)	2
06.2	<i>1 mark for each point.</i> • Preliminary study with widely spaced pH values to find where germination occurs (1) • Final range from below the lowest pH giving germination to above the highest, with closer intervals (1)	2
06.3	<i>1 mark for each point.</i> • Temperature (constant, natural germination temperature) (1) • Light, water volume, number of seeds per dish, seed source (1) • Growth medium (filter paper; soil would alter pH); nutrients not needed for germination (1) • Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure (1)	4
06.4	<i>1 mark for each point.</i> • Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured (1) • Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
07.1	<i>1 mark for each point.</i> • Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) (1) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people (1)	2
07.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Physical state of the effluent (solution / solid) (1) • Ocean currents / river flow (1) • Effect of pH and oxygen on solubility (1) • Bioaccumulation and food-chain concentration (1) • Food sources and consumption by local people (eg shellfish, seaweed) (1) • Half-lives of the isotopes (1) • Meteorological conditions for any gaseous release (1) • Nature of the environment - geology, vegetation (1)	4
07.3	<i>1 mark for each point.</i> • Monitoring focuses on the sites most at risk of contamination (1) • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions (1)	2
07.4	<i>1 mark for each point.</i> • It assesses the risk to the people most exposed because of their lifestyle before any health effects occur (1) • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
08.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) (1) • Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles (1) • Carbon neutral in principle: CO₂ released equals CO₂ absorbed in photosynthesis (1) • Energy density of alcohol / biodiesel nearly as high as fossil fuels; wastes (straw, sewage, landfill gas) are cheap (1)	3
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) (1) • Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO₂ as using fossil fuels directly (1) • Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally (1) • Supply from wastes limited to the amount produced (1)	3
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hydrogen from algae: some algae deprived of sulfur produce hydrogen during photosynthesis (1) • Anaerobic digestion extended to crop wastes and marine algae; small-scale digesters (eg one cow's manure supplying a family's cooking gas) (1) • Biofuels from microorganisms (eg oil-producing microalgae) (1)	2
08.4	<i>1 mark for each point.</i> • Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK (1) • New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	
09.1	<i>1 mark for each point.</i> • Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening (1) • Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight (1) • NO₂ (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O₂ to form ozone (1) • Ozone lags behind the NO / NO₂ peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight (1)	4
09.2	<i>1 mark for each point.</i> • NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure (1) • NO is oxidised in the atmosphere (by oxygen / ozone) to NO₂, a secondary pollutant (1)	2

Question	Marking guidance	Marks
09.3	1 mark for each point. • PANs (peroxyacetyl nitrates) (1) • Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x) (1)	2
09.4	Any 2 from the points below - 1 mark each, maximum 2. • Heavy vehicle use giving high NO_x and hydrocarbon emissions (1) • Sunny climate providing the UV for photochemical reactions (1) • Valley / basin topography and frequent temperature inversions trapping pollutants (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9

Question	Marking guidance	Marks
10.2	<i>1 mark.</i> The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts - Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) - Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects - Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) - Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) - Impacts before and during use; life-cycle / embodied energy - Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25